Chapter 4: Stochastic Calculus

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Abstract. *This section introduces a few selected topics from stochastic calculus—namely, Brownian motion, Ito processes, the Ito formula, Dynkin's formula, and Girsanov's theorem. Our aim is to cover only the essential material needed to formulate and analyze continuous-time stochastic optimal control problems in the subsequent sections. Comprehensive treatments of these topics can be found in standard references such as [1], [2], and [3].*

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4.1 Brownian Motion

Let us now consider the problem of modeling an $\mathbb{R}^n$-valued continuous-time random process X_t over the time interval $[0,T]$. Given a probability space $(\Omega, \mathcal{F}, \mathcal{P})$, such a random process can be defined as a function

$$X : [0,T] \times \Omega \to \mathbb{R}^n, \quad (t,\omega) \mapsto X_t(\omega). \tag{4.1}$$

For each $\omega \in \Omega$, the function $t \mapsto X_t(\omega)$ can be viewed as a sample path. For each $t \in [0,T]$, the function $\omega \mapsto X_t(\omega)$ is interpreted as the $\mathbb{R}^n$-valued random variable representing the value of the process at time t. For notational convenience, we usually write $X_t(\omega)$ simply as X_t.

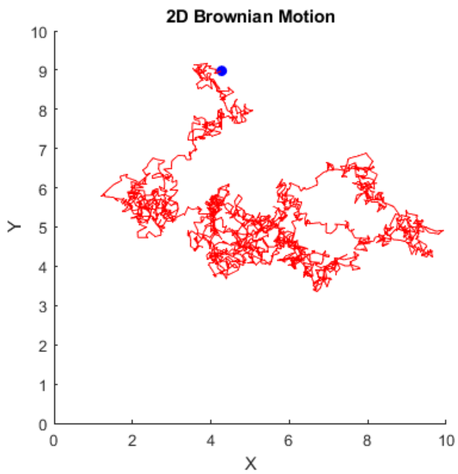

Figure 4.1: Random fluctuations of particles ejected from a pollen in water, first observed by the Scottish botanist Robert Brown in 1827.

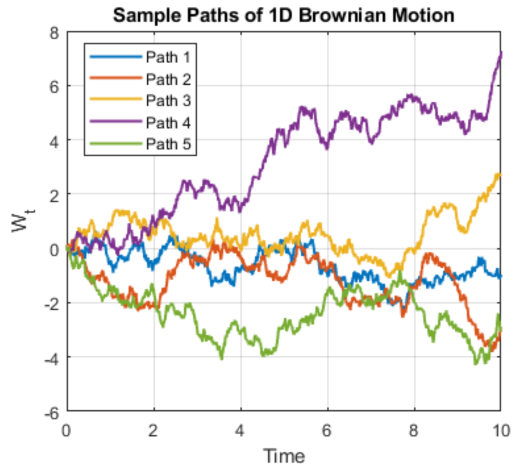

Figure 4.2: Simulated sample paths of the standard Brownian motion.

Unlike the discrete-time case, the rigorous treatment of continuous-time random processes is a far less trivial task, and its development required a century-long effort by physicists (including Einstein, Smoluchowski, Langevin, Ornstein, and Uhlenbeck) and mathematicians (including Wiener, Ito, and Stratonovich). In what follows, we begin with a mathematical characterization of *Brownian motion*, which serves as a fundamental building block for constructing more general stochastic processes. It is named after the Scottish botanist Robert Brown, who observed the restless and irregular motion of small particles suspended in liquids (Figure 4.1). However, its axiomatic characterization was developed much later by Norbert Wiener. Although the terms *Brownian motion* and *Wiener process* are now used synonymously, it is worth noting that nearly a century separates Brown's original observations [4, 5] and Wiener's foundational work [6].

Denote by $C([0,T],\mathbb{R}^n)$ the space of continuous functions $X : [0,T] \to \mathbb{R}^n$ with $X_0 = 0$, equipped with the Borel σ-algebra $\mathcal{B}(C([0,T],\mathbb{R}^n))$.[1] Given a probability space $(\Omega, \mathcal{F}, \mathcal{P})$, a random process $W_t(\omega), t \in [0,T]$ is called an n-dimensional Brownian motion if it satisfies the following properties:

(i) $W(\omega) \in C([0,T],\mathbb{R}^n)$ $\mathcal{P}$-almost surely. That is, for almost all $\omega \in \Omega$, the function $W(\omega) : [0,T] \to \mathbb{R}^n$ is continuous with $W_0 = 0$.

(ii) W_t has independent increments: for all $0 \leq t_1 < t_2 < \cdots < t_k \leq T$, the random variables $W_{t_1}, W_{t_2} - W_{t_1}, \ldots, W_{t_k} - W_{t_{k-1}}$ are independent.

(iii) The increments $W_t - W_s$ are Gaussian: $\mathbb{E}[W_t - W_s] = 0$ and $\mathbb{E}[(W_t - W_s)(W_t - W_s)^\top] = |t-s| I_n$, where I_n is the $n \times n$ identity matrix.

The existence of a random process satisfying the above conditions is guaranteed by Kolmogorov's continuity theorem [2, Theorem 2.2.3].

[1] An extensive treatment of Borel spaces can be found in [7].

Let $\mathcal{F}_t \subset \mathcal{F}$ be an increasing family of σ-algebras. (For example, $\mathcal{F}_t$ can be the σ-algebra generated by $\{W_s : 0 \leq s \leq t\}$.) We say that a random process X_t is *martingale* with respect to $\mathcal{F}_t$ if (i) X_t is $\mathcal{F}_t$-measurable $\forall t$; (ii) $\mathbb{E}[|X_t|] < \infty$ $\forall t$; and (iii) $\mathbb{E}[X_t|\mathcal{F}_s] = X_s$ for all $t \geq s$. It can be shown that Brownian motion W_t is a martingale with respect to the σ-algebra generated by itself $\{W_s : 0 \leq s \leq t\}$.

4.2 Stochastic Integrals

Recall that in discrete-time settings, stochastic state-space models were described using Borel-measurable stochastic kernels $\tau(x_{t+1} \mid x_t, u_t)$, or more explicitly as

$$X_{t+1} = F_t(X_t, U_t, W_t),$$

where W_t is a random variable representing an instantaneous stochastic disturbance. In this section, we introduce a continuous-time counterpart of such models. To this end, we need a mathematically rigorous framework to describe the evolution of a continuous-time random process $X_t,\ t \geq 0$, when it is causally affected by a Brownian motion $W_t,\ t \geq 0$. We adopt the concept of the Ito integral to model such an evolution in a precise way.

Since "white noise" can be loosely interpreted as the infinitesimal increments of Brownian motion, it is tempting to use an ordinary differential equation (ODE):

$$\frac{d}{dt}X_t = f(X_t) + g(X_t)\frac{d}{dt}W_t \tag{4.2}$$

to describe the stochastic evolution of X_t. Clearly, (4.2) makes sense only when the function W_t is differentiable in time, at least almost everywhere. However, this is a severe restriction, as it turns out that Brownian motion is nowhere differentiable almost surely [3, Section 2.9.D], and thus expressing white noise naively as $\frac{d}{dt}W_t$ is not justifiable. Therefore, to describe a "random process X_t driven by white noise," we must abandon an ODE-like representation (4.2) and seek an alternative formulation.

Let us rewrite (4.2) as the integral equation

$$X_t = x_0 + \int_0^t f_s(\omega)\, ds + \int_0^t g_s(\omega)\, dW_s. \tag{4.3}$$

Of course, (4.3) is only a formal reformulation of (4.2), and its meaning is not (yet) given. However, one can make the random process X_t well-defined by introducing proper *definitions* of the integrals in (4.3). It turns out that the first integral can be defined without much squabbling. Interpreting the second integral (called the *stochastic integral*), on the other hand, requires more subtle discussions. In fact, multiple interpretations are possible, leading to nonequivalent, yet self-consistent, paradigms of stochastic calculus. The two main paradigms that are widely adopted in today's literature are Ito's and Stratonovich's interpretations. The next two subsections summarize both interpretations. However, in later chapters, we will mostly adopt the Ito calculus.

4.3 The Ito Calculus

In this section, we consider how the SDE (4.3) is interpreted in the sense of Ito.

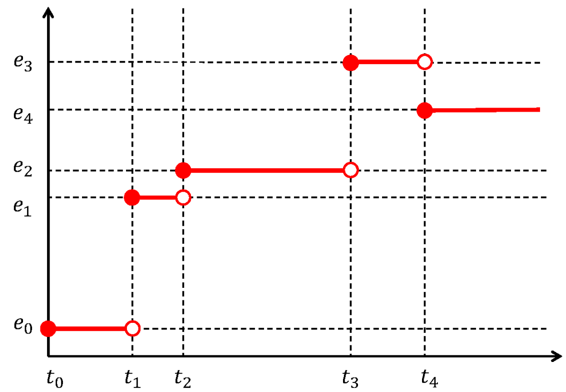

Figure 4.3: Example of an elementary function.

4.3.1 The Ito Integral

Assume all random processes are defined on a probability space $(\Omega, \mathcal{F}, \mathcal{P})$. For simplicity, consider the scalar case: X_t is an $\mathbb{R}$-valued process, W_t is a one-dimensional Brownian motion, and $f_t(\omega), g_t(\omega)$ are $\mathbb{R}$-valued processes whose sample paths depend on $\omega \in \Omega$. Let $\mathcal{F}_t \subset \mathcal{F}$ be an increasing family of σ-algebras such that W_t is a martingale with respect to $\{\mathcal{F}_t\}$.

The first integral in (4.3), $\int_0^t f_s(\omega)\, ds$, is well defined under the following conditions:

(i) $(t,\omega) \mapsto f_t(\omega)$ is $\mathcal{B}[0,\infty) \times \mathcal{F}$-measurable;

(ii) $f_t(\omega)$ is $\mathcal{F}_t$-adapted;

(iii) $\mathcal{P}\big(\int_0^t |f_s(\omega)|\, ds < \infty \ \text{ for all } t \geq 0\big) = 1$.

It turns out that the second integral in (4.3), $\int_0^t g_s(\omega)\, dW_s$ (called *the stochastic integral*), can also be defined unambiguously under the following conditions [2, Defs. 3.1.4, 3.3.2, 4.1.1]:

(i)' $(t,\omega) \mapsto g_t(\omega)$ is $\mathcal{B}[0,\infty) \times \mathcal{F}$-measurable;

(ii)' $g_t(\omega)$ is $\mathcal{F}_t$-adapted;

(iii)' $\mathcal{P}\big(\int_0^t g_s(\omega)^2\, ds < \infty \ \text{ for all } t \geq 0\big) = 1$.

The following is a brief explanation on how the *Ito integrals*, a particular type of stochastic integrals, can be defined under the conditions (i)'-(iii)'.

Let us start with the case in which $g_t(\omega)$ is a "simple" function. We say that a function $\phi(t,\omega)$ is *elementary* if it can be written as

$$\phi(t,\omega) = \sum_j e_j(\omega)\, \mathcal{X}_{[t_j, t_{j+1})}(t), \tag{4.4}$$

where $0 \leq t_0 < t_1 < \cdots < t_N \leq t$, $\mathcal{X}_{[t_j,t_{j+1})}$ is the indicator function of the interval $[t_j, t_{j+1})$, and $e_j(\omega)$ is $\mathcal{F}_{t_j}$-measurable. If $g_t(\omega)$ satisfies (i)'–(iii)' and is elementary, say $g_t(\omega) = \phi(t,\omega)$, then it is natural to define the stochastic integral by

$$\int_0^t g_s(\omega)\, dW_s := \sum_j e_j(\omega)\big(W_{t_{j+1}}(\omega) - W_{t_j}(\omega)\big). \tag{4.5}$$

Notice that this quantity is a random variable and depends on the sample path of W_t.

Next, we need to extend the definition of the stochastic integral (4.5) to non-elementary functions. To this end, we leverage the fact that for any g satisfying (i)'–(iii)', there exists a sequence of elementary functions $\{\phi_n\}$ such that

$$\mathbb{E}\left[\int_0^t \big(g_s(\omega) - \phi_n(s,\omega)\big)^2 \, ds\right] \to 0 \quad \text{as } n \to \infty. \tag{4.6}$$

Using this fact, the definition (4.5) can be generalized as follows:

Definition 4.1 (Ito integral) *Suppose g satisfies conditions (i)'–(iii)'. Let $\{\phi_n\}$ be a sequence of elementary functions such that* (4.6) *holds. The Ito integral of g from 0 to t is defined by*

$$\int_0^t g_s(\omega)\, dW_s(\omega) := \lim_{n\to\infty} \int_0^t \phi_n(s,\omega)\, dW_s(\omega), \tag{4.7}$$

where the limit converges in probability.

An important property, known as the Ito isometry, follows from this definition:

Lemma 4.1 (Ito isometry) *If g satisfies (i)'–(iii)', then for each $0 \le S \le T$,*

$$\mathbb{E}\left[\left(\int_S^T g_t(\omega)\, dW_t\right)^2\right] = \mathbb{E}\left[\int_S^T g_t(\omega)^2 \, dt\right]. \tag{4.8}$$

In summary, if $f_t(\omega)$ satisfies (i)–(iii) and $g_t(\omega)$ satisfies (i)'–(iii)', then the integral equation (4.3) is well defined. A process X_t represented by (4.3) is called an (one-dimensional) *Ito process*. Equation (4.3) is often written more compactly as

$$dX_t = f_t\, dt + g_t\, dW_t, \quad X_0 = x_0, \tag{4.9}$$

where x_0 is the initial condition. An equation of the form (4.9) is commonly referred to as a *stochastic differential equation (SDE)*.

Existence and uniqueness of solutions

While conditions (i)–(iii) and (i)'–(iii)' make the SDE (4.9) well defined, additional assumptions are required to ensure the existence and uniqueness of its solution. This is not surprising, since even for (deterministic) ordinary differential equations (ODEs), certain conditions are needed to guarantee existence and uniqueness (c.f., Picard-Lindelöf theorem). For example, the solution to $\dot{x}_t = x_t^2,\ x_0 = 0$ fails to exist on $[0,\infty)$, whereas the solution to $\dot{x}_t = \sqrt{x_t},\ x_0 = 0$ fails to be unique. For SDEs, several different notions of solutions exist, and the discussion of existence and uniqueness is correspondingly more subtle. See [2, Section 5.3] for a sufficient condition under which the existence and uniqueness of a *strong solution* are guaranteed.

Multidimensional Ito process

The discussion above generalizes to define the n-dimensional Ito process. Let $\{\mathcal{F}_t\}$ be an increasing family of σ-algebras such that an m-dimensional Brownian motion W_t is a martingale with respect to $\mathcal{F}_t$. Suppose $f_t(\omega) \in \mathbb{R}^{m\times 1}$ and $g_t(\omega) \in \mathbb{R}^{n\times m}$ are matrix-valued functions whose entries satisfy conditions (i)–(iii) and (i)'–(iii)', respectively. Then the integral equation (4.3) is well defined, and the process $X_t(\omega) \in \mathbb{R}^n$ defined by (4.3) is called an n-dimensional Ito process.

Numerical simulation

If an n-dimensional standard Brownian motion is to be simulated numerically with a constant step size Δt, one may use the scheme

$$W_{t+\Delta t} = W_t + \sqrt{\Delta t}\, V_t, \qquad V_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_n). \tag{4.10}$$

The standard deviation $\sqrt{\Delta t}$ of the additive noise is justified by property (iii) of Brownian motion described in Section 4.1.

More generally, the SDE (4.9) can be simulated with constant step size Δt via

$$X_{t+\Delta t} = X_t + f_t \Delta t + g_t \sqrt{\Delta t}\, V_t, \qquad V_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_n). \tag{4.11}$$

The scheme (4.11), known as the *Euler–Maruyama method*, is the natural generalization of the Euler method for ODEs.

Ito diffusion

A process X_t defined by the SDE

$$dX_t = \alpha_t(X_t)\, dt + \beta_t(X_t)\, dW_t, \qquad X_0 = x_0 \tag{4.12}$$

satisfies the Markov property.[2] Such a process is called an *Ito diffusion*.

4.3.2 The Ito Formula

Suppose that random processes X_t and Y_t are related by $Y_t = h(t, X_t)$ for some known mapping h. When X_t is an Ito process of the form

$$dX_t = f_t\, dt + g_t\, dW_t, \tag{4.13}$$

we are often interested in analyzing Y_t by deriving the stochastic differential equation that it satisfies. Assuming h is twice continuously differentiable, such an equation exists (so Y_t is also an Ito process), and it can in principle be derived directly from the definition of the Ito integral (Definition 4.1). However, this derivation is cumbersome and seldom used in practice.

Instead, one typically uses a simple set of rules, known as the *Ito formula*, which may be regarded as a generalization of the chain rule from elementary calculus. The Ito formula is summarized below.

[2] An $\mathcal{F}_t$-adapted process X_t is said to be *Markov* if $\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s]$ for all $0 \le s \le t$ and all bounded measurable f. Equivalently, X_t is Markov if $\mathbb{E}[f(X_t) \mid X_{t_1}, X_{t_2}, \ldots, X_{t_n}] = \mathbb{E}[f(X_t) \mid X_{t_n}]$ for all $0 \le t_1 \le t_2 \le \cdots \le t_n \le t$.

Theorem 4.1 *Let X_t be an $\mathbb{R}^n$-valued Ito process defined by* (4.13)*. Suppose $h(t,x)$ is twice continuously differentiable on $[0,\infty)\times\mathbb{R}^n$. Then $Y_t = h(t,X_t)$ is also an Ito process, and satisfies*

$$dY_t = \frac{\partial h}{\partial t}(t,X_t)\,dt + \left(\frac{\partial h}{\partial x}(t,X_t)\right)^\top dX_t + \frac{1}{2}\,\mathit{Tr}\!\left(\frac{\partial^2 h}{\partial x^2}(t,X_t)\,(dX_t)(dX_t)^\top\right), \tag{4.14}$$

where $\frac{\partial h}{\partial x}\in\mathbb{R}^n$ is the gradient (a column vector), $\frac{\partial^2 h}{\partial x^2}\in\mathbb{R}^{n\times n}$ is the Hessian matrix, and Tr$(\cdot)$ denotes the trace. Furthermore, the quadratic term $(dX_t)(dX_t)^\top$ can be simplified using the rules

$$(dt)^2 = 0, \quad (dt)(dW_t) = 0, \quad (dW_t)(dW_t)^\top = I_n\,dt. \tag{4.15}$$

Example 4.1 *Consider the n-dimensional Ito process* (4.13) *and define $Y_t = h(t,X_t)$ with $h(t,x) = \frac{1}{2}t^2\|x\|^2$. We compute*

$$\frac{\partial h}{\partial t} = t\|x\|^2, \quad \frac{\partial h}{\partial x} = t^2 x, \quad \frac{\partial^2 h}{\partial x^2} = t^2 I.$$

Applying the Ito formula gives

$$\begin{aligned} dY_t &= \frac{\partial h}{\partial t}(t,X_t)\,dt + \left(\frac{\partial h}{\partial x}(t,X_t)\right)^\top dX_t + \frac{1}{2}\,\mathit{Tr}\!\left(\frac{\partial^2 h}{\partial x^2}(t,X_t)(dX_t)(dX_t)^\top\right) \\ &= t\|X_t\|^2\,dt + t^2 X_t^\top(f_t\,dt + g_t\,dW_t) + \tfrac{1}{2}t^2(dX_t)^\top(dX_t). \end{aligned} \tag{4.16}$$

We also compute

$$\begin{aligned} (dX_t)^\top(dX_t) &= (f_t\,dt + g_t\,dW_t)^\top(f_t\,dt + g_t\,dW_t) \\ &= f_t^\top f_t(dt)^2 + 2f_t^\top g_t(dt)(dW_t) + \mathit{Tr}\big(g_t^\top g_t(dW_t)(dW_t)^\top\big) \\ &= \mathit{Tr}(g_t^\top g_t)\,dt, \end{aligned} \tag{4.17}$$

where the last step uses the rules (4.15)*. Combining* (4.16) *and* (4.17)*, we obtain*

$$dY_t = \Big(t\|X_t\|^2 + t^2X_t^\top f_t + \tfrac{1}{2}t^2\mathit{Tr}(g_t^\top g_t)\Big)\,dt + t^2X_t^\top g_t\,dW_t. \tag{4.18}$$

Exercise 4.1 *(Geometric Brownian motion) Consider the one-dimensional Ito process*

$$dX_t = aX_t\,dt + bX_t\,dW_t, \quad X_0 = X_{init} > 0. \tag{4.19}$$

Find the stochastic differential equation satisfied by $Y_t = \log X_t$. Find the solution to (4.19) *– that is, express X_t as an explicit function of X_{init}, t and W_t.*

Exercise 4.2 *Let W_t be an m-dimensional standard Brownian motion. Let $A\in\mathbb{R}^{n\times n}$ and $F\in\mathbb{R}^{n\times m}$ be constant matrices. Define an n-dimensional Ito process $X_t, t\geq 0$ by*

$$dX_t = AX_t\,dt + F\,dW_t, \tag{4.20}$$

and assume that $X_0\sim\mathcal{N}(0,\Sigma_0)$. Prove that $\Sigma_t = \mathbb{E}[X_tX_t^\top]$ satisfies the Lyapunov differential equation

$$\dot{\Sigma}_t = A\Sigma_t + \Sigma_t A^\top + FF^\top. \tag{4.21}$$

Exercise 4.3 *The Langevin equation for a particle of mass m with velocity V_t is given by*

$$dV_t = -\frac{\gamma}{m} V_t dt + F dW_t \tag{4.22}$$

where $-\gamma V_t$ is the friction force and W_t is the standard Brownian motion. The term $F dW_t$ represents the fluctuation force due to constant collisions of smaller molecules in the heat bath. Show that $F = \frac{\sqrt{2\gamma k T}}{m}$ is required if the temperature of the heat bath is T. Note that, by Boltzmann's equipartition theorem, the average kinetic energy $\frac{1}{2} m \mathbb{E}[V_t^2]$ per particle in the thermal equilibrium of temperature T should be equal to $\frac{1}{2} kT$, where k is the Boltzmann constant.

Using the Ito formula, one can also show that if X_t and Y_t are Ito processes, then

$$d(XY) = X\,dY + Y\,dX + dX\,dY. \tag{4.23}$$

We will use the following result in the proof of the Girsanov theorem in the next section.

Exercise 4.4 *Let $k(t,\omega) = [k_1(t,\omega) \ \cdots \ k_n(t,\omega)]^\top \in \mathbb{R}^n$ be a random process such that each $k_i(t,\omega)$, $i = 1, \ldots, n$, satisfies conditions (i)'–(iii)' on page 57. Define*

$$M_t(\omega) = \exp\left\{ \int_0^t k(s,\omega)^\top dW_s - \tfrac{1}{2} \int_0^t k(s,\omega)^\top k(s,\omega)\, ds \right\}. \tag{4.24}$$

Show that

$$dM_t = M_t\, k(t,\omega)^\top dW_t.$$

4.4 The Stratonovich integral

Let $\mathcal{P} = \{a = t_0 < t_1 < \cdots < t_N = b\}$ be a partition of the interval $[a,b] \subset \mathbb{R}$. The Riemann-Stieltjes integral of a real-valued function g with respect to another real-valued function W on the interval $[a,b]$ is defined to be

$$\int_a^b g_t dW_t = \lim_{N\to\infty} \sum_{i=0}^{N-1} g_{t_i'} (W_{t_{i+1}} - W_{t_i}) \tag{4.25}$$

where $t_i \le t_i' < t_{i+1}$ for each i.

If W_t is continuously differentiable, then (4.25) coincides with the Riemann integral $\int_a^b g_t \frac{dW_t}{dt} dt$, and the value of the integral does not depend on the choice of $t_i' \in [t_i, t_{i+1})$. In general, however, the value of (4.25) depends on the choice of the point t_i' at which the integrand g_t is evaluated. Specifically, if W_t is the Brownian motion (which is nowhere differentiable), the value of (4.25) depends on the choice of t_i'.

By (4.5) and Definition 4.1, it can be seen that the Ito integral is defined to be

$$\int_a^b g_t dW_t = \lim_{N\to\infty} \sum_{i=0}^{N-1} g_{t_i} (W_{t_{i+1}} - W_{t_i}). \tag{4.26}$$

That is, the integrand g is evaluated at $t_i' = t_i$, which is the right end of the interval $[t_i, t_{i+1})$.

On the other hand, the Stratonovich integral [8],[3] which leads to another widely used framework of stochastic calculus, is defined to be

$$\int_a^b g_t \circ dW_t = \lim_{N\to\infty} \sum_{i=0}^{N-1} g_{t_i'}(W_{t_{i+1}} - W_{t_i}) \tag{4.27}$$

where $t_i' = \frac{1}{2}(t_i + t_{i+1})$. That is, the integrand is evaluated at the midpoint of the interval $[t_i, t_{i+1})$.

While other choices of t_i' are possible, it turns out that $t_i' = t_i$ and $t_i' = \frac{1}{2}(t_i + t_{i+1})$ are the most sensible choices in view of their convenient mathematical properties. Pros and cons of each choice will be discussed at the end of this section.

4.4.1 Relationship between Ito and Stratonovich integrals

Suppose that an SDE is expressed using the Stratonovic integral as follows:

$$dX_t = \alpha^{(S)}(X_t, t)dt + \beta^{(S)}(X_t, t) \circ dW_t. \tag{4.28}$$

How can (4.28) be expressed using the Ito integral? Let

$$dX_t = \alpha^{(I)}(X_t, t)dt + \beta^{(I)}(X_t, t)dW_t \tag{4.29}$$

be the equivalent Ito SDE. [4] Let us derive relationships

$$\alpha^{(S)} \leftrightarrow \alpha^{(I)} \text{ and } \beta^{(S)} \leftrightarrow \beta^{(I)}$$

that will maintain the equivalence between (4.28) and (4.29). Introducing $\Delta t_i := t_{i+1} - t_i$, $\Delta X_{t_i} := X_{t_{i+1}} - X_{t_i}$ and $\Delta W_{t_i} := W_{t_{i+1}} - W_{t_i}$, the Stratonovich integral in (4.28) is defined[5] as

$$\int_a^b \beta^{(S)}(X_t, t) \circ dW_t := \lim_{N\to\infty} \sum_{i=1}^{N-1} \beta^{(S)}(X_{t_i} + \frac{1}{2}\Delta X_{t_i}, t_i)\Delta W_{t_i}. \tag{4.30}$$

Let us now evaluate the right-hand side of (4.30) using the Ito formula. From (4.29), we have

$$\Delta X_{t_i} = \alpha^{(I)}(X_{t_i}, t_i)\Delta t_i + \beta^{(I)}(X_{t_i}, t_i)\Delta W_{t_i} + O(\Delta t_i^{\frac{3}{2}}). \tag{4.31}$$

Notice that

$$\begin{aligned}
&\beta^{(S)}(X_{t_i} + \frac{1}{2}\Delta X_{t_i}, t_i) \\
&= \beta^{(S)}(X_{t_i}, t_i) + \frac{\partial \beta^{(S)}}{\partial x}\left(\frac{1}{2}\Delta X_{t_i}\right) + \frac{1}{2}\left(\frac{1}{2}\Delta X_{t_i}\right)^2 + O(\Delta t_i^{\frac{3}{2}}) \\
&= \beta^{(S)}(X_{t_i}, t_i) + \frac{1}{2}\frac{\partial \beta^{(S)}}{\partial x}\left(\alpha^{(I)}(X_{t_i}, t_i)\Delta t_i + \beta^{(I)}(X_{t_i}, t_i)\Delta W_{t_i}\right) \\
&\qquad\qquad + \frac{1}{8}\frac{\partial^2 \beta^{(S)}}{\partial x^2}\left(\beta^{(I)}(X_{t_i}, t_i)\right)^2 \Delta t_i + O(\Delta t_i^{\frac{3}{2}})
\end{aligned} \tag{4.32}$$

[3] We write $\int_a^b g_t \circ dW_t$ to denote the Stratonovich integral, whereas $\int_a^b g_t dW_t$ is assumed to mean the Ito integral.
[4] By *equivalent*, we mean that both (4.28) and (4.29) represent the same physical phenomenon, and that they provide consistent predictions about the statistical properties of the process X_t.
[5] The definition (4.30) is by Stratonovich [8]. The definitions (4.27) and (4.30) are equivalent – see [9, Thm 8.3.7].

Substituting this result into (4.30), and noticing that

$$\lim_{N\to\infty}\sum_{i=0}^{N-1}\beta^{(S)}(X_{t_i},t_i)\Delta W_{t_i}=\int_a^b\beta^{(S)}(X_t,t)dt \tag{4.33}$$

is the Ito integral, we obtain

$$\int_a^b\beta^{(S)}(X_t,t)\circ dW_t=\int_a^b\beta^{(S)}(X_t,t)dW_t+\frac{1}{2}\int_a^b\frac{\partial\beta^{(S)}}{\partial x}(X_t,t)\beta^{(I)}(X_t,t)dt. \tag{4.34}$$

Hence, (4.28) can be rewritten as an Ito SDE:

$$dX_t=\left[\alpha^{(S)}(X_t,t)dW_t+\frac{1}{2}\frac{\partial\beta^{(S)}}{\partial x}(X_t,t)\beta^{(I)}(X_t,t)\right]dt+\beta^{(S)}(X_t,t)dW_t. \tag{4.35}$$

Therefore, we find that if we choose $\alpha^{(I)}$ and $\beta^{(I)}$ to be

$$\alpha^{(I)}=\alpha^{(S)}+\frac{1}{2}\frac{\partial\beta^{(S)}}{\partial x}\beta^{(S)} \tag{4.36a}$$

$$\beta^{(I)}=\beta^{(S)} \tag{4.36b}$$

then (4.28) and (4.29) are equivalent. The (4.36) can be generalized to vector-valued processes. If $\alpha^{(S)},\alpha^{(S)}\in\mathbb{R}^{n\times 1}$ and $\beta^{(S)},\beta^{(S)}\in\mathbb{R}^{n\times m}$, then (4.28) and (4.29) are equivalent if

$$\alpha_i^{(I)}=\alpha_i^{(S)}+\frac{1}{2}\sum_{j,k}\beta_{kj}^{(S)}\frac{\partial\beta_{ij}^{(S)}}{\partial x_k} \tag{4.37a}$$

$$\beta^{(I)}=\beta^{(S)}. \tag{4.37b}$$

Conversely, for a given Ito SDE (4.29), the corresponding Stratonovich SDE can be constructed by

$$\alpha_i^{(S)}=\alpha_i^{(I)}-\frac{1}{2}\sum_{j,k}\beta_{kj}^{(I)}\frac{\partial\beta_{ij}^{(I)}}{\partial x_k} \tag{4.38a}$$

$$\beta^{(S)}=\beta^{(I)}. \tag{4.38b}$$

4.4.2 Chain rules

Let a Stratonovich SDE

$$dX_t=\beta(X_t,t)\circ dW_t \tag{4.39}$$

be given. If $f(x)$ is a function of x, what is the Stratonovich SDE for f?

In the Ito calculus, the Ito formula sets a particular "chain rule" for such a change of variables. In the Stratonovich calculus, a different (in fact, a simpler) chain rule applies.

The Taylor series expansion of a smooth function about a midpoint is given by

$$f(x+\frac{a}{2})-f(x-\frac{a}{2})=f'(x)a+\frac{f'''(x)}{3!}a^3+\cdots. \tag{4.40}$$

Since the quadratic term of a vanishes on the right-hand side,

$$f(X_{t_{i+1}}) = f(X_{t_i}) + f'(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\Delta X_{t_i} + O(\|\Delta X_{t_i}\|^3). \tag{4.41}$$

Notice that we have $O(\|\Delta X_{t_i}\|^3)$ instead of $O(\|\Delta X_{t_i}\|^2)$ by the virtue of evaluating f' at the midpoint.

From (4.39), we have

$$\Delta X_{t_i} = \beta(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\Delta W_{t_i} + O(\|\Delta W_{t_i}\|^3). \tag{4.42}$$

Substituting (4.42) into (4.41), and observing that both $O(\|\Delta X_{t_i}\|^3)$ and $O(\|\Delta W_{t_i}\|^3)$ diminish like $\Delta t_i^{\frac{3}{2}}$, we obtain

$$f(X_{t_{i+1}}) = f(X_{t_i}) + f'(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\beta(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\Delta W_{t_i} + O(\Delta t_i^{\frac{3}{2}}). \tag{4.43}$$

Noticing that

$$\lim_{N\to\infty}\sum_{i=0}^{N-1} f'(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\beta(X_{t_i} + \frac{1}{2}\Delta X_{t_i})\Delta W_{t_i} = \int_a^b f'(X_t)\beta(X_t)\circ dW_t \tag{4.44}$$

is the Stratonovich integral, (4.43) implies that

$$df(X_t) = \frac{\partial f}{\partial x}(X_t)\beta(X_t)\circ dW_t \tag{4.45}$$

is the Stratonovich SDE for f.

Remark 4.1 *The result* (4.45) *should be compared with the Ito formula. Recall that if $dX_t = \beta(X_t, t)dW_t$ is the Ito SDE, then the Ito SDE for $f(X_t)$ is given by*

$$\begin{aligned} df(X_t) &= \frac{\partial f}{\partial x}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(dX_t)^2 \\ &= \frac{\partial f}{\partial x}\beta(X_t)dW_t + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}\beta^2(X_t)dt. \end{aligned} \tag{4.46}$$

Comparing (4.45) *and* (4.46)*, we see that the Stratonovich scheme provides a simpler chain rule that looks more consistent with the chain rule in ordinary calculus. This is a significant advantage of the Stratonovich integral over the Ito integral.*

It is easy to generalize the chain rule (4.45) to a multi-dimensional SDE of the form

$$dX_t = \alpha(X_t, t)dt + \beta(X_t, t)\circ dW_t. \tag{4.47}$$

If the function $f(x, t)$ is differentiable in each variable, then the Stratonovich SDE for $Y_t = f(X_t, t)$ is

$$\begin{aligned} dY_t &= \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial x}dX_t \\ &= \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial x}\alpha\right)dt + \frac{\partial f}{\partial x}\beta\circ dW_t \end{aligned} \tag{4.48}$$

where $\frac{\partial f}{\partial t}$ and $\frac{\partial f}{\partial x}$ are interpreted as the gradient matrices.

Exercise 4.5 *The conditions* (4.37) *and* (4.38) *imply that the coefficients of Ito and Stratonovich SDEs coincide if the diffusion coefficient g does not depend on x. In general, however, it must always be clarified in what sense (e.g., Ito or Stratonovich) a given SDE should be interpreted. The reference [10] emphasizes this point using a simple geometric Brownian motion.*

(i) In Exercise 4.1, the solution to the Ito SDE

$$dX_t = aX_t dt + bX_t dW_t, \quad X_0 = X_{init} > 0 \tag{4.49}$$

was derived. Show that $\mathbb{E}[X_t] \to 0$ as $t \to \infty$ if $a < 0$.

(ii) Show that $\bar{X}_t = X_{init} \exp(\bar{a}t + \bar{b}W_t)$ solves the Stratonovich SDE

$$d\bar{X}_t = \bar{a}\bar{X}_t dt + \bar{b}\bar{X}_t \circ dW_t, \quad X_0 = X_{init} > 0 \tag{4.50}$$

(iii) Show that $\mathbb{E}[\bar{X}_t] \to 0$ as $t \to \infty$ if $\bar{a} + \frac{1}{2}\bar{b}^2 < 0$.

(iv) Show that the results from (i) and (iii) do not contradict. Hint: Use (4.37) *and* (4.38)*.*

4.4.3 Ito vs Stratonovich integrals

As we have already seen in Remark 4.1, the Stratonovich integral offers the simpler chain rule that looks consistent to the chain rule in ordinary calculus. This is in contrast to the Ito formula, which indicates that a somewhat special formula is required in Ito calculus. This argument supports the reason why Stratonovich integrals are sometimes preferred over Ito integrals.

Another argument that supports the Stratonovich integral is based on the "white noise limit" argument [11, Ch.8]. As we saw in Section 4.2 already, an ODE representation like (4.2) is not valid for non-differentiable signals W_t like Brownian motions. However, it is reasonable to consider a sequence of random processes $W_t^{(n)}$ with continuously differentiable sample paths such that

$$W_t^{(n)}(\omega) \to W_t(\omega) \text{ as } n \to \infty$$

uniformly in $t \in [a, b]$ for almost all $\omega \in \Omega$. Let $X_t^{(n)}(\omega)$ be the solution of

$$\frac{d}{dt}X_t = f(X_t) + g(X_t)\frac{d}{dt}W_t^{(n)}. \tag{4.51}$$

It can be shown that there exists a function $X_t(\omega)$ such that

$$X_t^{(n)}(\omega) \to X_t(\omega) \text{ as } n \to \infty$$

uniformly in $t \in [a, b]$ for almost all $\omega \in \Omega$. Crucially, it turns out (see [2, Sec. 3.3],[12] and references therein) that X_t coincides with the solution of the Stratonovich SDE with the same coefficients:

$$dX_t = f(X_t) + g(X_t) \circ dW_t.$$

This argument strongly favors the Stratonovich integral over other possible choices of stochastic integrals.

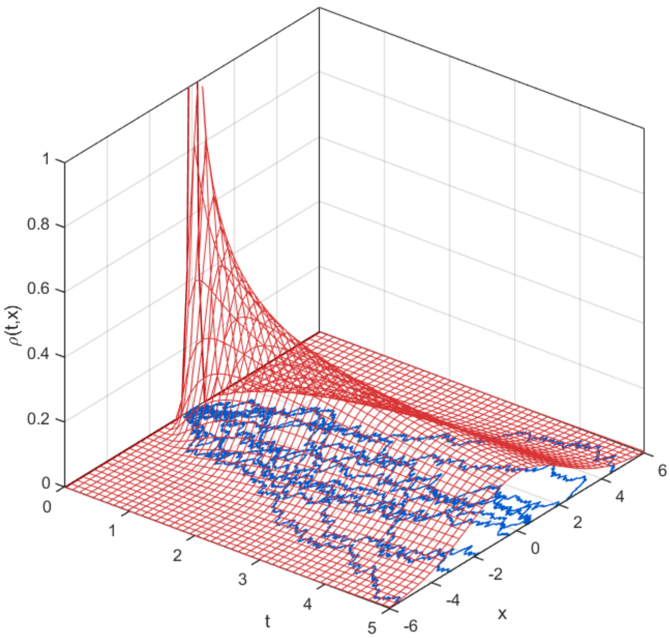

Figure 4.4: Diffusion of standard Brownian motion and the corresponding solution to the Fokker–Planck equation.

However, the Ito integral has a clear advantage in one aspect: It is a martingale, i.e., $\mathbb{E}\left[\int_a^b g_t dW_t\right] = 0$, by construction. In contrast, $\mathbb{E}\left[\int_a^b g_t \circ dW_t\right] \neq 0$ in general. This difference comes from the different choices of $t_i' \in [t_i, t_{i+1})$ at which the integrand is evaluated. Ito's choice $t_i' = t_i$ makes g_t non-anticipatory, ensuring the Ito integral is a martingale.

Remark 4.2 *(Ito vs Stratonovich controversy) ([2, Section 3.3], [13] [10])*

4.5 Fokker-Planck Equation

As a simple application of the Ito formula, let us derive the one-dimensional Fokker-Planck equation. The Fokker-Planck equation is a partial differential equation (PDE) that describes how the probability density function $\rho(t, x)$ with respect to the variable x evolves over time t when x follows the Ito diffusion

$$dX_t = \alpha(X_t)\, dt + \beta(X_t)\, dW_t, \qquad \rho(0, \cdot) \sim \rho_0. \tag{4.52}$$

If (4.52) is scalar, $\rho(t, x)$ can be visualized as a two-dimensional surface. See Figure 4.4 for the solution to the Fokker-Planck equation corresponding to standard Brownian motion. In the following, we derive the Fokker-Planck equation associated with the one-dimensional diffusion process (4.52) using the Ito formula.

Consider an arbitrary C^2 test function[6] $\psi(x)$ with compact support $[a,b] \subset \mathbb{R}$. By the Ito formula, we have

$$\begin{aligned} d\psi(X_t) &= \frac{\partial \psi}{\partial x}(X_t)\, dX_t + \frac{1}{2}\frac{\partial^2 \psi}{\partial x^2}(X_t)\, (dX_t)^2 \\ &= \frac{\partial \psi}{\partial x}(X_t)\big(\alpha(X_t)\, dt + \beta(X_t)\, dW_t\big) + \frac{1}{2}\frac{\partial^2 \psi}{\partial x^2}(X_t)\, \beta^2(X_t)\, dt. \end{aligned} \tag{4.53}$$

Taking expectations with respect to the PDF $\rho(t,x)$ gives

$$d\mathbb{E}[\psi(X_t)] = \mathbb{E}\left[\frac{\partial \psi}{\partial x}(X_t)\alpha(X_t) + \tfrac{1}{2}\frac{\partial^2 \psi}{\partial x^2}(X_t)\beta^2(X_t)\right] dt. \tag{4.54}$$

Since (4.54) does not involve stochastic integrals, it may be rewritten as

$$\frac{d}{dt}\mathbb{E}[\psi(X_t)] = \mathbb{E}\left[\frac{\partial \psi}{\partial x}(X_t)\alpha(X_t) + \tfrac{1}{2}\frac{\partial^2 \psi}{\partial x^2}(X_t)\beta^2(X_t)\right]. \tag{4.55}$$

The left-hand side of (4.55) can be expressed as

$$\frac{d}{dt}\int_a^b \psi(x)\rho(t,x)\, dx = \int_a^b \psi(x)\, \frac{\partial \rho}{\partial t}(t,x)\, dx. \tag{4.56}$$

The right-hand side of (4.55) becomes

$$\int_a^b \left[\frac{\partial \psi}{\partial x}(x)\alpha(x) + \tfrac{1}{2}\frac{\partial^2 \psi}{\partial x^2}(x)\beta^2(x)\right]\rho(t,x)\, dx \tag{4.57a}$$

$$\begin{aligned} &= \underbrace{\big[\psi(x)\alpha(x)\rho(t,x)\big]_a^b}_{0} - \int_a^b \psi(x)\, \frac{\partial}{\partial x}(\alpha(x)\rho(t,x))\ dx \\ &\qquad + \tfrac{1}{2}\underbrace{\big[\tfrac{\partial \psi}{\partial x}(x)\beta^2(x)\rho(t,x)\big]_a^b}_{0} - \tfrac{1}{2}\int_a^b \frac{\partial \psi}{\partial x}(x)\, \frac{\partial}{\partial x}\big(\beta^2(x)\rho(t,x)\big)\ dx \end{aligned} \tag{4.57b}$$

$$\begin{aligned} &= -\int_a^b \psi(x)\, \frac{\partial}{\partial x}(\alpha(x)\rho(t,x))\ dx - \tfrac{1}{2}\underbrace{\big[\psi(x)\tfrac{\partial}{\partial x}\big(\beta^2(x)\rho(t,x)\big)\big]_a^b}_{0} \\ &\qquad + \tfrac{1}{2}\int_a^b \psi(x)\, \frac{\partial^2}{\partial x^2}\big(\beta^2(x)\rho(t,x)\big)\ dx \end{aligned} \tag{4.57c}$$

$$= \int_a^b \psi(x)\left[-\frac{\partial}{\partial x}\big(\alpha(x)\rho(t,x)\big) + \tfrac{1}{2}\frac{\partial^2}{\partial x^2}\big(\beta^2(x)\rho(t,x)\big)\right] dx. \tag{4.57d}$$

Since (4.56) must equal (4.57d) for every test function ψ with compact support and every interval $[a,b]$, we conclude

$$\frac{\partial \rho}{\partial t}(t,x) = -\frac{\partial}{\partial x}(\alpha(x)\rho(t,x)) + \tfrac{1}{2}\frac{\partial^2}{\partial x^2}\big(\beta^2(x)\rho(t,x)\big). \tag{4.58}$$

[6] That is, ψ has two continuous derivatives.

Equation (4.58) is the *Fokker–Planck equation*.

More generally, let

$$dX_t = \alpha(X_t)\,dt + \beta(X_t)\,dW_t, \qquad X_0 \sim \rho_0, \tag{4.59}$$

be an n-dimensional Ito diffusion. Then, the multidimensional Fokker-Planck equation is given by

$$\frac{\partial \rho}{\partial t}(t,x) = -\sum_i \frac{\partial}{\partial t}\left[\alpha_i(x)\rho(t,x)\right] + \frac{1}{2}\sum_{i,j}\left[\beta_i(x)\beta_j^\top(x)\rho(t,x)\right] \tag{4.60}$$

where $\beta_i(x)$ is the i-th row of the matrix $\beta(x)$.

Exercise 4.6 *(Brownian bridge) Let W_t be the standard Brownian motion. For fixed $a, b \in \mathbb{R}$, consider the SDE:*

$$dX_t = \frac{b - X_t}{1-t}dt + dW_t, \quad 0 \le t < 1, \quad X_0 = a. \tag{4.61}$$

The process defined by (4.61) *is called the* Brownian bridge. *Figure 4.5 shows* 20 *sample paths of the Brownian bridge process* (4.61).

(i) Show that the process

$$X_t = a(1-t) + bt + (1-t)\int_0^t \frac{dW_s}{1-s}, \quad 0 \le t < 1 \tag{4.62}$$

solves the SDE (4.61). *Prove that $X_t \to b$ as $t \to 1$ almost surely.*

(ii) Assume $a = b = 0$ for simplicity. Show that the Fokker-Planck equation corresponding to the SDE (4.61) *is*

$$\frac{\partial \rho}{\partial t}(t,x) = \frac{1}{1-t}\rho(t,x) + \frac{x}{1-t}\frac{\partial \rho}{\partial x}(t,x) + \frac{1}{2}\frac{\partial^2 \rho}{\partial x^2}(t,x) \tag{4.63}$$

for $0 < t < 1$, $x \in \mathbb{R}$.

(iii) Show that if $\rho_1(t,x)$ solves the Fokker-Planck equation (4.63)*, then $\rho_2(t,x) := \rho_1(1-t,x)$ also solves the same Fokker-Planck equation* (4.63).

4.5.1 Forward and backward Kolmogorov equations

Consider the n-dimensional Ito diffusion

$$dX_t = \alpha(X_t)\,dt + \beta(X_t)\,dW_t, \qquad t_0 \le t \le t_1 \tag{4.64}$$

with the initial condition $X_{t_0} = x_0$. Let $\rho(t,x)$ be the solution to the corresponding Fokker-Planck equation (4.60) with the initial condition $\rho(t_0,x) = \delta_{x_0}(x)$. To indicate its initial condition explicitly, let us denote the solution by $p(t,x|t_0,x_0) := \rho(t,x)$. The Kolmogorov forward equation for the diffusion process (4.64) is a partial differential equation (PDE) for $p(t,x|t_0,x_0)$ viewed as a

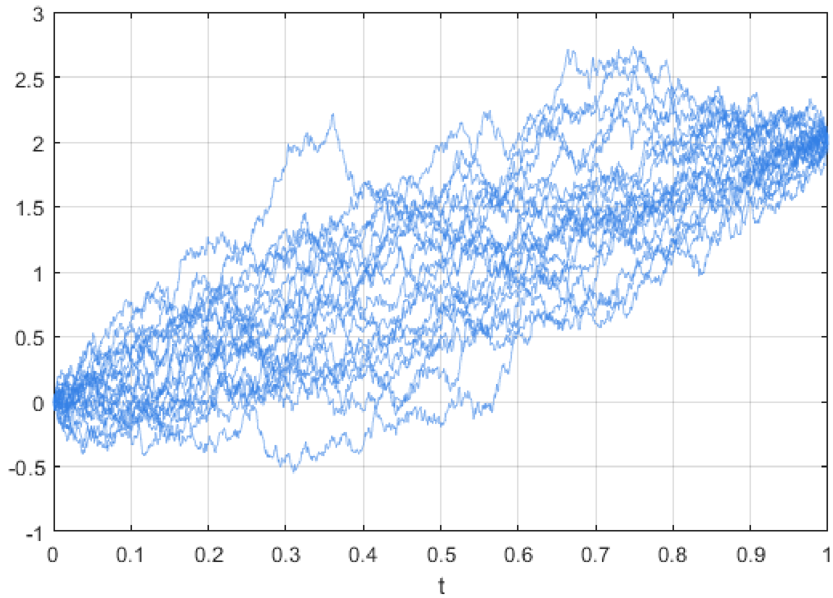

Figure 4.5: Sample paths of the Brownian bridge process (4.61) with $a = 0$ and $b = 2$.

function of (t, x), for a given initial condition (t_0, x_0). Clearly, the Kolmogorov forward equation is equivalent to the Fokker-Planck equation (4.60). That is,

$$\frac{\partial}{\partial t}p(t,x|t_0,x_0) = -\sum_i \frac{\partial}{\partial t}[\alpha_i(x)p(t,x|t_0,x_0)] + \frac{1}{2}\sum_{i,j}\frac{\partial}{\partial t}[\beta_i(x)\beta_j^\top(x)p(t,x|t_0,x_0)] \tag{4.65}$$

with the initial condition $p(t, x|t_0, x) = \delta_{x_0}(x)$.

If (t_1, x_1) is fixed, the conditional probability distribution $p(t_1, x_1|t, x)$ can also be viewed as a function of the initial state (t, x). The Kolmogorov backward equation is a PDE satisfied by $p(t_1, x_1|t, x)$, viewed as a function (t, x), for a given terminal state (t_1, x_1). Since we already know the Fokker-Planck equation (equivalently, the Kolmogorov forward equation), deriving the backward equation is relatively easy. In what follows, we provide a derivation for the scalar case.

Let us select an arbitrary distribution $\rho(t, x)$ at time $t < t_1$. For each $s \in [t, t_1]$, let $\rho(s, x)$ be the solution of the Fokker-Planck equation where $\rho(t, x)$ serves as the initial condition. Since $p(t_1, x_1|t, x)$ is a conditional probability distribution, we must have

$$\begin{aligned}\rho(t_1,x_1) &= \int_{\mathcal{X}} p(t_1,x_1|t,x)\rho(t,x)dx \\ &= \int_{\mathcal{X}} p(t_1,x_1|s,x)\rho(s,x)dx.\end{aligned} \tag{4.66}$$

In particular, if we define $f(s) := \int_{\mathcal{X}} p(t_1, x_1|s, x)\rho(s, x)dx$ as a function of s, it takes a constant value for all $s \in [t, t_1]$. Therefore, $f'(t) = \lim_{s\searrow t}\frac{f(s)-f(t)}{s-t} = 0$, or equivalently

$$\frac{d}{dt}f(t) = \int_{\mathcal{X}} \frac{\partial}{\partial t}(p\rho)dx = 0 \tag{4.67}$$

is required. Assuming that $\rho(\pm\infty) = 0$ and $\frac{d}{dt}\rho(\pm\infty) = 0$, the quantity above can be computed as follows:

$$\int_{\mathcal{X}} \left[\frac{\partial p}{\partial t}\rho + p\frac{\partial \rho}{\partial t} \right] dx = \int_{\mathcal{X}} \left[\frac{\partial p}{\partial t}\rho - p\frac{\partial}{\partial t}(\alpha\rho) + \frac{1}{2}p\frac{\partial^2}{\partial t^2}(\beta^2 r) \right] dx \tag{4.68}$$

$$= \int_{\mathcal{X}} \left[\frac{\partial p}{\partial t} + \alpha\frac{\partial p}{\partial t} + \frac{1}{2}\beta^2\frac{\partial^2 p}{\partial t^2} \right] \rho dx \tag{4.69}$$

$$\tag{4.70}$$

The first equality (4.68) follows from the Fokker-Planck equation $\frac{\partial \rho}{\partial t} = -\frac{\partial}{\partial x}(\alpha\rho) + \frac{1}{2}\frac{\partial^2}{\partial x^2}(\beta^2\rho)$. The second line (4.69) is obtained via integration by parts. Since the last quantity (4.69) must be zero for an arbitrary choice of ρ, we conclude that the following Kolmogorov backward equation must hold:

$$\frac{\partial}{\partial t}p(t_1, x_1|t, x) = -\alpha(x)\frac{\partial}{\partial x}p(t_1, x_1|t, x) - \frac{1}{2}\beta^2(x)\frac{\partial^2}{\partial x^2}p(t_1, x_1|t, x). \tag{4.71}$$

This is a PDE defined on the interval $t \in [t_0, t_1]$. It can be solved backward in time with the terminal condition $p(t_1, x_1|t_1, x) = \delta_{x_1}(x)$.

If x is $\mathbb{R}^n$-valued, the Kolmogorov backward equation is generalized to

$$\frac{\partial}{\partial t}p(t_1, x_1|t, x) = -\sum_i \alpha_i(x)\frac{\partial}{\partial x_i}p(t_1, x_1|t, x) - \frac{1}{2}\sum_{i,j}\beta_i(x)\beta_j^\top(x)\frac{\partial^2}{\partial x_i \partial x_j}p(t_1, x_1|t, x). \tag{4.72}$$

4.5.2 Generator of an Ito Diffusion

For the one-dimensional Ito diffusion (4.52) with $X_0 = x$ and a C^2 function ψ, the operator A defined by

$$A\psi(x) := \lim_{t \searrow 0} \frac{\mathbb{E}[\psi(X_t)] - \psi(x)}{t} \tag{4.73}$$

is called the *(infinitesimal) generator* of (4.52). From (4.55), we see that

$$A\psi(x) = \alpha(x)\frac{\partial \psi}{\partial x}(x) + \tfrac{1}{2}\beta^2(x)\frac{\partial^2 \psi}{\partial x^2}(x). \tag{4.74}$$

More generally, let

$$dX_t = \alpha(X_t)\, dt + \beta(X_t)\, dW_t, \qquad X_0 = x, \tag{4.75}$$

be an n-dimensional Ito diffusion with

$$\alpha(x) = \begin{bmatrix} \alpha_1(x) \\ \vdots \\ \alpha_n(x) \end{bmatrix}, \qquad \beta(x) = \begin{bmatrix} \beta_{11}(x) & \cdots & \beta_{1m}(x) \\ \vdots & & \vdots \\ \beta_{n1}(x) & \cdots & \beta_{nm}(x) \end{bmatrix}.$$

For a C^2 function $\psi : \mathbb{R}^n \to \mathbb{R}$, the generator (4.73) of (4.75) is given by

$$A\psi(x) = \sum_i \alpha_i(x)\frac{\partial \psi}{\partial x_i}(x) + \tfrac{1}{2}\sum_{i,j}\left[\beta(x)\beta^\top(x)\right]_{ij}\frac{\partial^2 \psi}{\partial x_i \partial x_j}(x). \tag{4.76}$$

4.6 Dynkin's formula

Let $\mathcal{F}_t \subset \mathcal{F}$ an increasing family of σ-algebras. A function $\tau : \Omega \to [0, \infty)$ is called *stopping time with respect to* $\mathcal{F}_t$ if $\{\omega : \tau(\omega) \leq t\} \in \mathcal{F}_t, \ \forall t \geq 0.$ Intuitively, $\tau(\omega)$ is a stopping time if it is possible to decide whether or not the event $\tau(\omega) \leq t$ has occurred on the basis of the knowledge of $\mathcal{F}_t$.

Example 4.2 *Let $\mathcal{F}_t$ be the σ-algebra generated by $\{X_s : s \leq t\}$. Then, the first exit time*

$$\tau_{\mathcal{X}} := \inf\{t > 0 : X_t \notin \mathcal{X}\}$$

is a stopping time. However, $\tau'_{\mathcal{X}} := \inf\{t > 0 : X_{2t} \notin \mathcal{X}\}$ is not a stopping time since such a function is not $\mathcal{F}_t$ measurable.

The next theorem provides Dynkin's formula, which can be thought of as a generalization of the *fundamental theorem of calculus*:

$$f(\tau) = f(0) + \int_0^\tau f'(t)dt \tag{4.77}$$

for a differentiable function f.

Theorem 4.2 *(Dynkin's formula) Let*

$$dX_t = \alpha(X_t)dt + \beta(X_t)dW_t, \ \ X_0 = x \tag{4.78}$$

be an Ito diffusion, and let $\mathcal{F}_t$ be the σ-algebra generated by $\{W_s : s \leq t\}$. Suppose

- *$f : \mathbb{R}^n \to \mathbb{R}$ is C^2 and has compact support;*
- *τ is a stopping time with respect to $\mathcal{F}_t$ such that $\mathbb{E}[\tau] < \infty$.*

Assume that X_t belongs to the support of f. Then

$$\mathbb{E}[f(X_\tau)] = f(x) + \mathbb{E}\left[\int_0^\tau Af(X_s)ds\right] \tag{4.79}$$

where

$$Af(X_s) = \sum_i \alpha_i(X_s)\frac{\partial f}{\partial x_i}(X_s) + \frac{1}{2}\sum_{i,j}\left[\beta(X_s)\beta^\top(X_s)\right]_{ij}\frac{\partial^2 f}{\partial x_i \partial x_j}(X_s). \tag{4.80}$$

Proof: Put $Y_t = f(X_t)$ and apply Ito's formula:

$$dY_t = \sum_i \frac{\partial f}{\partial x_i}dX_t + \frac{1}{2}\sum_{i,j}\frac{\partial^2 f}{\partial x_i \partial x_j}dX_{i,t}dX_{j,t} \tag{4.81a}$$

$$= \sum_i \alpha_i \frac{\partial f}{\partial x_i}dt + \frac{1}{2}\sum_{i,j}\frac{\partial^2 f}{\partial x_i \partial x_j}(\beta dW_t)_i(\beta dW_t)_j + \sum_i \frac{\partial f}{\partial x_i}(\beta dW_t)_i \tag{4.81b}$$

$$= \sum_i \alpha_i \frac{\partial f}{\partial x_i}dt + \frac{1}{2}\sum_{i,j}(\beta\beta^\top)_{ij}\frac{\partial^2 f}{\partial x_i \partial x_j}dt + \sum_{i,k}\beta_{ik}\frac{\partial f}{\partial x_i}dW_{k,t}. \tag{4.81c}$$

Therefore,

$$
\begin{aligned}
f(X_t) &= f(x) \\
&+ \int_0^\tau \left\{ \sum_i \alpha_i \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i,j} (\beta\beta^\top)_{ij} \frac{\partial^2 f}{\partial x_i \partial x_j} \right\} ds + \sum_{i,k} \int_0^\tau \beta_{ik} \frac{\partial f}{\partial x_i} dW_{k,s}. \qquad (4.82)
\end{aligned}
$$

Taking the expectation of both sides, we obtain (4.79). ■

Example 4.3 *(The graph of Brownian motion) Let W_t be the one-dimensional standard Brownian motion. Define a two-dimensional random process $\bar{X}_t$ by*

$$
\begin{cases} \bar{X}_{1,t} = dt, & X_{1,0} = 0 \\ \bar{X}_{2,t} = dW_t, & X_{2,0} = 0 \end{cases} \tag{4.83}
$$

which can be written more compactly as

$$
d\bar{X}_t = \alpha \bar{X}_t dt + \beta dW_t \tag{4.84}
$$

with $\alpha = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\beta = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$. Since $\bar{X}_{1,t} = t$, the first coordinate can be interpreted as time. Hence, a sample path of (4.84) *may be regarded as a graph of Brownian motion.*

Let $f(t, w)$ be a C^2 function $\mathbb{R}^2 \to \mathbb{R}$. By Theorem 4.2, for each stopping time τ we have

$$
\mathbb{E}[f(\tau, W_\tau)] = f(0, 0) + \mathbb{E}\left[\int_0^\tau A f(s, W_s) ds \right] \tag{4.85}
$$

where

$$
A f(t, w) = \frac{\partial f}{\partial t} + \frac{1}{2} \frac{\partial^2 f}{\partial w^2}. \tag{4.86}
$$

If $f(t, w)$ does not depend on w, we have $\frac{\partial^2 f}{\partial w^2} = 0$ and (4.85) *reduces to the fundamental theorem of calculus* (4.77).

The argument of Example 4.3 can be generalized to an n-dimensional Ito diffusion

$$
dX_t = f_t(X_t) dt + \sigma_t(X_t) dW_t, \quad X_0 = x_0. \tag{4.87}
$$

A graph of X is an $(n + 1)$-dimensional process defined by

$$
\begin{cases} \bar{X}_{1,t} = dt, & X_{1,0} = t_0 \\ \bar{X}_{2,t} = dX_t, & X_{2,0} = x_0. \end{cases} \tag{4.88}
$$

Let $f(t, x)$ be a C^2 function $\mathbb{R}^{n+1} \to \mathbb{R}$. By applying Dynkin's formula to (4.88), we obtain

$$
\mathbb{E}[f(\tau, X_\tau)] = f(t_0, x_0) + \mathbb{E}\left[\int_{t_0}^\tau A f(s, X_s) ds \right] \tag{4.89}
$$

where

$$
A f(t, X_t) = \frac{\partial f}{\partial t} + f_t^\top(X_t) \frac{\partial f}{\partial x} + \frac{1}{2} \mathrm{Tr} \left[\sigma_t(X_t) \sigma_t^\top(X_t) \frac{\partial^2 f}{\partial x^2} \right] \tag{4.90}
$$

In (4.90), $\frac{\partial f}{\partial x}$ is a (column) gradient vector and $\frac{\partial^2 f}{\partial x^2}$ is a symmetric Hessian matrix. The result (4.89) will be used in the derivation of the stochastic Hamilton-Jacobi-Bellman (HJB) equation in the next chapter.

4.7 The Girsanov theorem

In Section 4.1, we saw that if W_t is Brownian motion, then (i) it is a martingale, i.e., $\mathbb{E}[W_t \mid \mathcal{F}_s^W] = W_s$ for $t \geq s$, and (ii) $\mathbb{E}[(W_t - W_s)(W_t - W_s)^\top \mid \mathcal{F}_s^W] = (t-s)I_n$ for $t \geq s$, where $\mathcal{F}_s^W$ is the σ-algebra generated by $\{W_u : 0 \leq u \leq s\}$. In a certain sense, the converse also holds: any continuous, square-integrable martingale satisfying (i), (ii), and $W_0 = 0$ is (necessarily) standard Brownian motion. The following is Lévy's characterization of Brownian motion [2, 1].

Theorem 4.3 *Let $(\Omega, \mathcal{F}, \mathcal{P})$ be a probability space and $\{\mathcal{F}_t\} \subset \mathcal{F}$ a nondecreasing family of σ-algebras. Let $W : \Omega \to C([0,T], \mathbb{R}^n)$ be an $\{\mathcal{F}_t\}$-adapted process with $W_0 = 0$ such that, for $0 \leq s \leq t \leq T$,*

$$\mathbb{E}[W_t \mid \mathcal{F}_s^W] = W_s \quad \mathcal{P}\text{-a.s.}, \tag{4.91}$$

$$\mathbb{E}[(W_t - W_s)(W_t - W_s)^\top \mid \mathcal{F}_s^W] = (t-s)I_n \quad \mathcal{P}\text{-a.s.}. \tag{4.92}$$

Then W_t is standard Brownian motion.

Recall that Section 4.1 also introduced independent increments and joint Gaussianity. Theorem 4.3 shows these properties *follow* solely from (4.91)–(4.92).

Lemma 4.2 *[2, Lemma 8.6.2] Let μ and ν be probability measures on $(\Omega, \mathcal{F})$ with*

$$d\nu(\omega) = f(\omega)\, d\mu(\omega)$$

for some $f \in L^1(\mu)$. Let X be a random variable with $\mathbb{E}_\nu(|X|) = \int_\Omega |X| f \, d\mu < \infty$. For a sub-$\sigma$-algebra $\mathcal{G} \subset \mathcal{F}$,

$$\mathbb{E}_\mu[fX \mid \mathcal{G}] = \mathbb{E}_\nu[X \mid \mathcal{G}]\, \mathbb{E}_\mu[f \mid \mathcal{G}]. \tag{4.93}$$

Proof: For each $G \in \mathcal{G}$,

$$\begin{aligned}
\int_G \mathbb{E}_\mu[fX|\mathcal{G}]d\mu &= \int_G X f d\mu && (4.94a)\\
&= \int_G X d\nu && (4.94b)\\
&= \int_G \mathbb{E}_\nu[X|\mathcal{G}]d\nu && (4.94c)\\
&= \int_G \mathbb{E}_\nu[X|\mathcal{G}]f d\mu && (4.94d)\\
&= \int_\Omega \mathcal{X}_G \mathbb{E}_\nu[X|\mathcal{G}]f d\mu && (4.94e)\\
&= \mathbb{E}_\mu[\mathcal{X}_G \mathbb{E}_\nu[X|\mathcal{G}]f] && (4.94f)\\
&= \mathbb{E}_\mu[\mathbb{E}_\mu[\mathcal{X}_G \mathbb{E}_\nu[X|\mathcal{G}]f|\mathcal{G}]] && (4.94g)\\
&= \mathbb{E}_\mu[\mathcal{X}_G \mathbb{E}_\nu[X|\mathcal{G}]\mathbb{E}_\mu[f|\mathcal{G}]] && (4.94h)\\
&= \int_G \mathbb{E}_\nu[X|\mathcal{G}]\mathbb{E}_\mu[f|\mathcal{G}]d\mu. && (4.94i)
\end{aligned}$$

Equalities (4.94a) and (4.94c) hold by definition of conditional expectation, whereas (4.94a) and (4.94d) follows from (4.93). In step (4.94e), we introduced the indicator function $\mathcal{X}_G$. We used the definition of expectation in (4.94f) and the law of iterated expectations in (4.94g). The equation (4.94h) follows from the fact that $\mathbb{E}[XY|\mathcal{G}] = X\mathbb{E}[Y|\mathcal{G}]$ if X is $\mathcal{G}$-measurable [2, Theorem B.2]. This establishes (4.93). ■

Theorem 4.4 *(The Girsanov theorem, [2, Theorem 8.6.3] [1, Theorem 6.3]) Let $(\Omega, \mathcal{F}, P)$ be a probability space. Let W_t, $0 \leq t \leq T$, be an n-dimensional Brownian motion under P, and let $\mathcal{F}_t = \sigma\{W_s : 0 \leq s \leq t\}$. Suppose $Y_t \in \mathbb{R}^n$ is an Ito process of the form*

$$dY_t = K_t(\omega)\, dt + dW_t, \qquad Y_0 = 0,$$

where K_t is $\mathcal{F}_t$-adapted function such that

$$\mathbb{E}_P\left[\exp\left(\tfrac{1}{2}\int_0^T \|K_t\|^2 dt\right)\right] < \infty.$$

Define a probability measure Q on $(\Omega, \mathcal{F}_T)$ by $dQ(\omega) = M_t(\omega)dP(\omega)$ where

$$M_t(\omega) = \exp\left(-\int_0^t K_s^\top(\omega)dW_s - \frac{1}{2}\int_0^t \|K_s(\omega)\|^2 ds\right), \quad 0 \leq t \leq T. \tag{4.95}$$

Then Y_t, $0 \leq t \leq T$, is an n-dimensional Brownian motion under Q.

Proof: By Lévy's characterization (Theorem 4.3), it is sufficient to show that

(i) $\mathbb{E}\left[Y_t|\mathcal{F}_s\right] = Y_s \quad Q - a.s., \quad 0 \leq s \leq t;$

(ii) $\mathbb{E}\left[(Y_t - Y_s)(Y_t - Y_s)^\top|\mathcal{F}_s\right] = (t-s)I \quad Q - a.s., \quad 0 \leq s \leq t.$

To show (i), first recall that from the result of Exercise 4.4, we have $dM_t = -M_t K_t^\top dW_t$. In particular, M_t is a martingale with respect to P, i.e., $\mathbb{E}_P\left[M_t|\mathcal{F}_s\right] = M_s$. Moreover, if we set $N_t = M_t Y_t$, using (4.23),

$$\begin{aligned}
dN_t &= Y_t dM_t + M_t dY_t + dM_t dY_t \\
&= Y_t(-M_t K_t^\top dW_t) + M_t(K_t dt + dW_t) + (-M_t K_t^\top dW_t)(K_t dt + dW_t) \\
&= M_t(I - Y_t K_t^\top)dW_t.
\end{aligned}$$

Hence, N_t is a martingale with respect to P, i.e., $\mathbb{E}_P\left[M_t Y_t|\mathcal{F}_s\right] = M_s Y_s$. Therefore, by Lemma 4.2,

$$\mathbb{E}_Q\left[Y_t|\mathcal{F}_s\right] = \frac{\mathbb{E}_P\left[M_t Y_t|\mathcal{F}_s\right]}{\mathbb{E}_P\left[M_t|\mathcal{F}_s\right]} = \frac{M_s Y_s}{M_s} = Y_s. \tag{4.96}$$

This proves (i). To show (ii), it is sufficient to prove

$$\mathbb{E}\left[(Y_t - Y_s)(Y_t - Y_s)^\top - (t-s)I|\mathcal{F}_s\right] = 0.$$

Set $Z_t = (Y_t - Y_s)(Y_t - Y_s)^\top - (t-s)I$. Similarly to part (i), it can be shown that $M_t Z_t$ is a martingale, i.e., $\mathbb{E}\left[M_t Z_t | \mathcal{F}_s\right] = M_s Z_s$. Therefore,

$$\mathbb{E}_Q\left[Z_t|\mathcal{F}_s\right] = \frac{\mathbb{E}_P\left[M_t Z_t|\mathcal{F}_s\right]}{\mathbb{E}_P\left[M_t|\mathcal{F}_s\right]} = \frac{M_s Z_s}{M_s} = Z_s = 0. \tag{4.97}$$

This completes the proof. ■

Theorem 4.4 shows that the probability measure Q is absolutely continuous with respect to P ($Q \ll P$), and the Radon-Nikodym derivative $\frac{dQ}{dP}$ is given by

$$\frac{dQ}{dP}(\omega) = M_T(\omega). \tag{4.98}$$

Noticing that $M_T(\omega) > 0$ almost surely, we also have $P \ll Q$ and

$$\begin{aligned}\frac{dP}{dQ}(\omega) &= M_T^{-1}(\omega) &\text{(4.99a)}\\ &= \exp\left(\int_0^T K_s^\top(\omega)dW_s(\omega) + \frac{1}{2}\int_0^T \|K_s(\omega)\|^2 ds\right) &\text{(4.99b)}\\ &= \exp\left(\int_0^T K_s^\top(\omega)dY_s(\omega) - \frac{1}{2}\int_0^T \|K_s(\omega)\|^2 ds\right) &\text{(4.99c)}\end{aligned}$$

where in the last step we used $dY_s = K_s\,ds + dW_s$.

Remark 4.3 *A useful interpretation of* (4.98) *and* (4.99) *is as follows. Suppose we generate a sample path $W(\omega)$ of Brownian motion under P, and form Y via*

$$dY_t(\omega) = K_t(\omega)\,dt + dW_t(\omega).$$

Under P, Y is generally not *Brownian unless $K \equiv 0$. However, according to Girsanov's theorem, Y_t is a Brownian motion with respect to a different probability measure Q provided $\frac{dQ}{dP}(\omega) = M_T(\omega)$. This means that it is possible to observe the sample path $Y(\omega)$ as a realization of a Brownian motion. In fact, observing $Y(\omega)$ as an outcome of this experiment (this occurs with probability $\propto P(\omega)$) is M_T^{-1} times as likely as observing the same sample path $Y(\omega)$ as a realization of a Brownian motion (this occurs with probability $\propto Q(\omega)$).*

To formalize Remark 4.3, let $C([0,T],\mathbb{R}^n)$ be the space of continuous functions $x : [0,T] \to \mathbb{R}^n$ with $x_0 = 0$, equipped with the Borel σ-algebra $\mathcal{B}(C([0,T],\mathbb{R}^n))$. Define probability measures on this path space by

$$\mu_Y(B) = P\{\omega : Y(\omega) \in B\}, \quad \mu_W(B) = P\{\omega : W(\omega) \in B\}$$

for each $B \in \mathcal{B}(C([0,T],\mathbb{R}^n))$. Then

$$\mu_W(B) = Q\{Y \in B\} \tag{4.100a}$$
$$= \int_{\{\omega: Y \in B\}} dQ(\omega) \tag{4.100b}$$
$$= \int_{\{\omega: Y \in B\}} \frac{dQ}{dP}(\omega)\, dP(\omega) \tag{4.100c}$$
$$= \int_{\{\omega: Y \in B\}} M_T(\omega)\, dP(\omega) \tag{4.100d}$$
$$= \int_B M_T(y)\, d\mu_Y(y). \tag{4.100e}$$

Theorem 4.5 *[1, Theorem 7.11] Let $(\Omega, \mathcal{F}, P)$ be a complete probability space and $\mathcal{F}_t \subset \mathcal{F}$, $0 \le t \le T$ a nondecreasing family of σ-algebras. Suppose W_t, $0 \le t \le T$ is an $\mathcal{F}_t$-measurable Brownian motion, Y_t, $0 \le t \le T$ is an $\mathcal{F}_t$-measurable random process such that $\mu_Y \sim \mu_W$, and $\mathcal{F}_t^Y$ is the σ-algebra generated by Y_s, $0 \le s \le t$. Then, there exists an $\mathcal{F}_t^Y$-measurable Brownian motion $\hat{W}_t$ and an $\mathcal{F}_t^Y$-measurable random process K_t such that*

$$Y_t = \int_0^t K_s(\omega) ds + \hat{W}_t \tag{4.101}$$
$$P\left(\int_0^T \|K_s(\omega)\|^2 ds < \infty\right) = 1. \tag{4.102}$$

4.8 Historical remarks

The Brownian motion was named after Robert Brown (1773-1858), a Scottish botanist, who observed incessant and irregular motion of small particles ejected from the pollen of the plant *Clarkia pulchella* in water using a one-lens microscope in 1827 [4, 5]. While Brown was not the first to observe such a random walk of small particles in a fluid, he conducted a series of experiments using particles of glass, minerals, and a drop of water of microscopic size immersed in oil, which refuted an early hypothesis that the cause of the motion was life-related. However, it took several decades until a unanimous view was formed on the exact cause of the Brownian motion.

In the early 20th century, Brownian motion was of central interest among theoretical and experimental physicists, including Albert Einstein [14], Marian Smoluchowski [15], and William Sutherland [16], who pursued the atomic-molecular view of matter. Although Einstein was initially unaware of Brown's work, he predicted the existence of such a phenomenon purely based on theoretical grounds. His main motivation for studying this subject was to find solid evidence of the existence of atoms of definite, finite size, which was still a subject of debate at that time. To this end, he considered small particles of a microscopically visible size suspended in liquids, and characterized the osmotic pressure from the viewpoint of molecular-kinetic theory. Based on several statistical assumptions, he derived a formula that relates the diffusion coefficient of the Brownian particles and Avogadro's number (or equivalently, the size of atoms). Physicists first attempted to measure the velocity of a Brownian particle directly to determine its kinetic energy. However, they soon realized the major difficulty of such an approach due to the very irregular and nowhere differentiable nature of the particle's trajectory. To avoid this difficulty, an experimental physicist, Jean Perrin [17], observed colloidal dispersions under the influence of gravity. From this experiment, he identified the diffusion coefficient, from which he also obtained Avogadro's number, which agreed to within 19% of the modern value. Perrin's result convinced even the last skeptics, including Wilhelm Ostwald, to accept the existence of atoms and molecules. The agreement of Perrin's result and Max Planck's determination of the molecular size from the black-body radiation further solidified the atomic-molecular picture of matter.

The roles that Brownian motion played in the early formation of the molecular-kinetic theory are well-documented in [18, 19].

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